CLAIMS

What is claimed is:

1. A data storage apparatus, comprising:

- a non-volatile storage medium;
- a storage controller communicatively coupled to the storage medium; and
- a massively parallel optical interface, coupled to the storage controller, configured to serve as the primary data input/output port for the apparatus.

2. The apparatus of claim 1, wherein the optical interface comprises a two-dimensional array of photon emitters and a two-dimensional array of photon sensors.

3. The apparatus of claim 1, wherein the optical interface replaces a serial electrical interface such as SATA, SAS, or NVMe.

4. The apparatus of claim 1, wherein the apparatus is configured to transfer data via the optical interface as a sequence of light frames at a rate exceeding 100 gigabits per second.

5. A storage system, comprising a host computer and at least one Optically-Attached Storage (OAS) apparatus according to claim 1, wherein the host computer and the OAS apparatus are connected via a massively parallel optical link.